Time Series Decomposition

Trend · Seasonality · Residuals

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Learning objectives

By the end of this chapter, you will be able to:

- write down the additive and multiplicative decomposition models for a time series;
- estimate the **trend** by moving average, parametric (polynomial) and non-parametric (LOESS) regression;
- estimate the **seasonality** by seasonal averaging, dummy regression, and cosine–sine (Fourier) models;
- produce a deseasonalised, detrended residual and check it for stationarity (preview of Chapter 3).

Chapter outline

1. The decomposition model (additive vs multiplicative)
2. Trend estimation (moving average, parametric, LOESS)
3. Seasonality estimation (averaging, dummy, Fourier)
4. Residuals and diagnostics
5. Full decomposition workflow

The decomposition model

1. Additive and multiplicative models

Definition – Two classical decompositions

$$\text{Additive:} \qquad Y_t = T_t + S_t + \varepsilon_t,$$

$$\text{Multiplicative:} \qquad Y_t = T_t \cdot S_t \cdot \varepsilon_t.$$

Here T_t is the trend, S_t the seasonality (period d), and ε_t a stationary residual.

- Additive model: seasonal amplitude is roughly *constant* over time.
- Multiplicative model: amplitude *grows with the level* ($Y_t > 0$); take $\log Y_t$ to recover the additive form.
- Goal: estimate $\widehat{T}_t$ and $\widehat{S}_t$, then $\widehat{\varepsilon}_t = Y_t - \widehat{T}_t - \widehat{S}_t$ should look stationary.

Trend estimation

2. Symmetric (centred) moving average

Definition – Moving average

For odd window $2k+1$ (with k small relative to T):

$$\widehat{T}_t = \frac{1}{2k+1} \sum_{i=-k}^{k} Y_{t+i}, \quad t = k+1, \dots, T-k.$$

Properties.

- **Preserves linear trends:** if $Y_t = a + b\,t$, then $\widehat{T}_t = a + b\,t$ (exact).
- **Removes additive periodic seasonality with period $d = 2k+1$.**
- Loses k observations at each end (boundary effect).

2. Numerical example: $k = 1$ (3-point MA)

Series $Y = (10, 12, 15, 14, 16, 18, 20)$:

$$
\begin{aligned}
\widehat{T}_2 &= \tfrac{10+12+15}{3} \approx 12.33, \\
\widehat{T}_3 &= \tfrac{12+15+14}{3} \approx 13.67, \\
\widehat{T}_4 &= \tfrac{15+14+16}{3} = 15.00, \\
\widehat{T}_5 &= \tfrac{14+16+18}{3} = 16.00, \\
\widehat{T}_6 &= \tfrac{16+18+20}{3} = 18.00.
\end{aligned}
$$

Boundaries $t = 1$ and $t = 7$ are not estimated ($k = 1$ data point lost on each side).

2. Parametric trend (polynomial regression)

Definition – Polynomial trend

$$T_t \;=\; \beta_0 + \beta_1\, t + \beta_2\, t^2 + \cdots + \beta_p\, t^p.$$

- $p = 1$: straight line. Estimate (β_0, β_1) by OLS.
- $p = 2$: quadratic. Captures curvature.
- p large : risk of overfitting and bad extrapolation.

Choose p via AIC / BIC, cross-validation or domain knowledge. Refrain from polynomials of order > 3 — they are unstable at boundaries.

2. Linear trend in Python (OLS)

```
import numpy as np
from sklearn.linear_model import LinearRegression

t = np.arange(1, len(y) + 1).reshape(-1, 1)
model = LinearRegression().fit(t, y)
trend = model.predict(t)
beta0, beta1 = model.intercept_, model.coef_[0]
```

Reading. $\widehat{\beta}_1$ is the per-period slope of the trend. Subtract `trend` from `y` to detrend.

2. Non-parametric trend (LOESS / splines)

Definition – Local polynomial / LOESS

For each t, fit a low-degree polynomial *locally* on a window around t with weights decreasing in $|t' - t|$ (typically a tricube kernel). The window width is a hyper-parameter (*span*).

- Flexibility: handles non-linear, non-parametric trends.
- Sensitive to boundary effects (less than MA but still).
- Other choices: **splines** (penalised cubic), **wavelets**.

Hyper-parameter trade-off: small span $\Rightarrow$ overfitting, large span $\Rightarrow$ over-smoothing.

2. LOESS in Python

```
from statsmodels.nonparametric.smoothers_lowess import lowess

trend = lowess(y, t, frac=0.2, return_sorted=False)

# detrended series
detrended = y - trend
```

frac $=$ fraction of data in each local window ($\sim$ smoothing parameter).

Seasonality estimation

3. Seasonal averaging (after detrending)

Once the trend is removed, observations at the same season k ($k = 1, \ldots, d$) are averaged:

$$\widehat{S}_k = \frac{1}{n_k} \sum_{t \in \text{season } k} (Y_t - \widehat{T}_t).$$

- For monthly data: $d = 12$, one $\widehat{S}_k$ per calendar month.
- Convention: re-centre so $\sum_k \widehat{S}_k = 0$ (additive case).
- Deseasonalised series: $Y_t^{\text{adj}} = Y_t - \widehat{S}_{k(t)}$.

3. Dummy-variable regression

Definition – Seasonal dummies

Define $D_{k,t} = 1$ if t is in season k, 0 otherwise.

$$Y_t \ = \ \beta_0 + \sum_{k=2}^{d} \beta_k \, D_{k,t} + \varepsilon_t .$$

Estimated by OLS. Use $d - 1$ dummies (drop one to avoid collinearity).

- Equivalent to seasonal averaging when no other regressor is present.
- Easy to combine with a polynomial trend in the same regression.

3. Cosine–sine (Fourier) model

Definition – Trigonometric seasonality

With period d and angular frequency $\omega = 2\pi/d$:

$$S_t = a\cos(\omega t) + b\sin(\omega t).$$

Higher harmonics: add $\cos(j\omega t)$, $\sin(j\omega t)$ for $j = 2, 3, \ldots, \lfloor d/2 \rfloor$.

- **Compact:** 2 parameters per harmonic vs. $d - 1$ dummies.
- Generalises to non-integer or non-periodic seasonal patterns.
- With *all* harmonics, exactly equivalent to dummies (linear span argument).

3. Fourier seasonality in Python

```
import numpy as np
import statsmodels.api as sm

omega = 2 * np.pi / 12
X = np.column_stack([np.cos(omega * t), np.sin(omega * t)])
X = sm.add_constant(X)

model = sm.OLS(y, X).fit()
season = model.predict(X)
```

Add a column `t` to `X` for a simultaneous linear trend.

Residuals and diagnostics

4. Residuals after decomposition

$$\widehat{\varepsilon}_t \;=\; Y_t - \widehat{T}_t - \widehat{S}_t.$$

Quality checks.

- **No remaining trend:** plot $\widehat{\varepsilon}_t$ versus t; should fluctuate around 0.
- **No remaining seasonality:** plot the autocorrelation (ACF) — no spike at lag d.
- **Constant variance:** no funnel shape; otherwise ARCH-type effect (Chapter 6).
- **Approximate normality:** histogram + QQ-plot.

The residual $\widehat{\varepsilon}_t$ is the input to ARMA modelling in Chapter 5.

Workflow

5. Full decomposition workflow

1. Plot Y_t. Decide additive vs. multiplicative (constant or growing seasonal amplitude).
2. If multiplicative, work on $\log Y_t$.
3. Estimate $\widehat{T}_t$ (MA, polynomial, or LOESS).
4. Detrend: $\widetilde{Y}_t = Y_t - \widehat{T}_t$.
5. Estimate $\widehat{S}_t$ from $\widetilde{Y}_t$.
6. Residual: $\widehat{\varepsilon}_t = Y_t - \widehat{T}_t - \widehat{S}_t$.
7. Diagnostics on $\widehat{\varepsilon}_t$.
8. If residual is stationary: model with ARMA (Chapter 5).

Up next

Chapter 3. Stationary time series.

- Strict and weak stationarity.
- Autocovariance function and ACF/PACF.
- White noise.
- Wold decomposition.

Thank you.

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